

Tracking the Blast Wave of a Supernova - What Can We Learn About SN2017ein and its Progenitor by Modelling its Environment

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In this research we attempt to model the circumstellar medium/environment around the radio driven type Ic supernova SN2017ein and compare physical parameters to similar types of supernovae (type Ib, type Ic and radio type II). This is done by applying radio modelled theory including broken power law fits on data received from the very large array assuming synchrotron self absorption dominating the radiation in which we observe from an approximated disc in the sky showing the supernova and its blast wave. We calculate radio comparable parameters $B = 0.20 \pm 0.03$ G, $R = 2.1 \pm 0.5 \times 10^{16}$ cm, $\dot{M} = 9.4 \pm 3.0 \times 10^{10}$ g/cm and $\rho_{CSM} = 1.7 \pm 1.0 \times 10^{-23}$ g/cm³. Then by comparing these values to other radio supernovae we conclude our data implies SN2017ein is a relatively low density radio faint supernova and that it was caused by a binary stripping scenario involving a Wolf-Rayet star with a companion. Our results are not completely conclusive due to the assumptions and simplifications done in the time frame which are explored in thorough detail.

I. Introduction

A. Overall Aims

In this project, the overall aim is to accurately model the environment around a

stellar explosion of a dying star known as a supernova, focusing specifically on the supernova SN2017ein. We will do this by looking at images of the sky containing these objects in different wavelengths (in

this project we are working on radio wavelength data), performing analysis and using fitting methods on the data in the images to extract physical parameters about the environments around the stellar explosions (formally known as the CSM - Circumstellar Medium). This is extremely useful particularly as we physically constrain properties (e.g. density) about the environment we can conduct several studies including finding out information about the progenitor star that caused the explosion and compare our results to other similar supernovae modelling and conclude properties about SN2017ein. Specifically, we will compute and model the physical parameters for the radio supernova SN2017ein, and then compare it to similar radio supernovae (Radio SNe) to attempt to constrain properties about how it compares to others and its progenitor star.

B. Overview of Theory Used

In this project, to understand the methods used and context of results we will need to start by diving into what supernovae are, how they are caused and how they are classified since SN2017ein is a certain type of supernova belonging to the class of type Ic (more will be discussed on classes in the next section). Since radio telescopes have been used to collect the data, it is useful to also explore how a radio telescope

array (like the very large array or VLA) works and what the images we see in our software actually are especially since our data was collected by the very large array. This then naturally leads us into a discussion about the software used to clean our images and account for issues caused when radio telescopes are used and then fitting these images with 2D Gaussian Ellipses to actually find physical brightness values. Then the modelling starts, with our brightness (flux) values we can then create a spectral energy distribution (SED) which is a measure of flux against frequency (we already know the frequency of each image). We can use physical models to fit the SED for useful parameters which will be used to extract physical parameters about the supernova. Then to actually get these physical parameters for the supernova we need a model to work with. There is already extensive framework and modelling used for radio SNe and we will repeat what [1, 2] did to model the environment/CSM around the explosion and its forward moving blast wave sent out from it. This theory/framework will now be explored in detail.

II. Theory and Methods

A. Theory

A.1 Background on Supernovae

It is very well understood that stars are powered by the nuclear fusion of hydrogen in their core. When a star completely empties this supply of hydrogen (or when enough material is accreted onto them by a binary companion) and if it is massive enough (roughly above 8 solar masses) the compression from its gravity (since no outward pressure from fusion remains) can cause a huge stellar explosion - a supernova. This is because the compression reaches extremely compact densities with huge velocities and due to physical density limits (like electron degeneracy pressure) a shock wave is sent back outwards when the absolute density limit is reached. Then even heavier stars can have such strong collapses that after their explosion they compress into a neutron star or black hole. [3] explores how the mass of the star which causes the supernova (the progenitor) can link to its final product.

There are many different types of supernovae which are classified based on the material observed in their spectra (emission or absorption lines observed which vary depending on materials present). In this project, we are focusing on modeling a type Ic supernovae (2017ein). Type Ic SNe are

observed without hydrogen or helium in their spectra. This means that they are caused from a stripped envelope core collapse process where both the hydrogen and helium layers were removed before the explosion usually via stellar winds or binary (two stars orbiting a shared centre of gravity) interactions. Stellar winds are usually formed when materials in outer layers of the star gain enough energy to escape the star's gravitational field, and binary interactions cause stripping because one star in the system can expand to the point it reaches its Roche lobe limit (a limit to the size of the star depending on the binary system's gravitational interaction) and then overflow of materials begin which can lead to unstable mass transfer, resulting in the outer layers of a star being transferred away. Because hydrogen fuses into helium, the two outermost layers will be hydrogen and then helium, so any sort of stripping mechanism will remove these first creating what we call a Wolf-Rayet (WR) star. The supernovae themselves are powered by the radioactive decay of Nickel-56 releasing approximately 10^{44} J of energy.

Due to there being no hydrogen or helium, a strong classifier for the progenitors of a type Ic SN are classical WR stars which have masses ranging from 8 to 30 solar masses and are usually in the helium burning phase meaning they are at the end of their life cycle as a star. They have

broad and strong emission lines (He II, N, C, O) formed in powerful, fast radioactively driven winds with outflow accelerated to thousands of km/s. This outflow is what distinguishes the WR stars from massive helium stars [4].

A..2 SN2017ein

We will now explore the supernova SN2017ein in more detail as this is the object we are focusing on modeling in our research. This paper by Kilpatrick et al [5] reports that the first credible detection of a type Ic progenitor is SN2017ein. This is very important as it could explain the mechanics behind radio driven supernovae and constrain what kind of objects cause them. Pre-explosion Hubble Space Telescope imaging and post-explosion Keck adaptive optics precisely located the progenitor candidate which the best match suggests it was a massive 80 solar mass star stripped by a binary companion which caused the explosion (hydrogen and helium were stripped in the binary interaction). By comparing the photometry (measures of brightness) against evolution models for single stars and binaries, the data showed SN2017ein resembled a carbon-rich, low luminosity (luminosity is a measure of brightness) SN Ic. This paper also suggested these SN Ic progenitor stars may come from binary systems with WR stars, not just sin-

gle WRs.

Xiang et al [6] also explore this supernova, they constrained properties about SN2017ein through shock breakout modelling and again pre-explosion HST images. The results align very well with the Kilpatrick et al paper as the modelling suggested relatively low ejecta mass which reinforced the binary stripping scenario (and also the spectra showed it is carbon rich - consistent with a stripped progenitor).

So what is actually known and still up for speculation about SN2017ein? It occurred in the spiral galaxy NGC 3938 with a distance of 18.9 ± 2.6 Mpc (from [6]), calculated from the Tully Fisher relation - a distribution which models the distance to galaxies based on mass constraints. Xiang et al [6] also observed supernova emission relative to the Nickel decay model very early at one day after the explosion. They calculated the residual envelope for the compact circumstellar medium/ CSM (the environment/material around the supernova which can be modelled to infer physical properties about it or the progenitor) to have a radius $R \approx 8 \pm 4$ solar radii and $M \approx 0.02$ solar masses. This argues for a low-density, tiny CSM and disfavours a single WR star progenitor. While there are no direct measurements of the CSM (very difficult as it is very small and has low density), estimates of the Nickel-56 mass in the CSM give an approximate value of 0.037 so-

lar masses, an approximate ejected mass of 1.246 solar masses and an energy output of around 1.641×10^{44} J [7]. There are no direct measurements of the wind velocity yet (which is a huge limitation for us that we will explore later) but in general WR-like stars can adopt solar wind speeds of 10^3 km/s to 3×10^3 km/s [8].

While the [5] and [6] both imply a binary system as the progenitor, this question still is not fully answered and confirmed. Van Dyke et al [9] identified a point source in archival HST data that may correspond to the progenitor, but Zhao et al [10] actually found this source had not disappeared and suggested the candidate is a random unrelated star closer to SN2017ein’s position. Another discrepancy is that the ejecta mass is quite low compared to an expected Wolf-Rayet progenitor (1-2 solar masses vs 10s of solar masses).

A.3 Radio Telescopes/Astronomy

Our data was collected from the VLA so we will explore how radio telescopes and astronomy work in general to understand how our data was collected. Radio astronomy studies celestial objects by detecting and analysing radio waves they emit. Unlike visible light, radio waves can pass through dust and can be probed in day and night and do not require space-based telescopes. For example we previously explored how [2]

used the VLA, which in more detail is an array of telescopes synchronised with each other to observe the supernova SN2004C.

Radio telescopes are instruments designed to detect radio frequency radiation. They have very large parabolic reflector dishes as radio waves have the largest wavelengths and some sort of system which converts faint radio waves into electrical voltages. Then electronics are used to digitize and convert these voltages into computable signals. You can have single-dish instruments (e.g. the Green Bank Telescope) or interferometric arrays like the VLA.

In this project, we make use of data received from the VLA hence it is crucial to understand how the interferometry performed by the telescope array works. Interferometry is very useful because it allows us to infer spatial information about the objects measured and a famous example is Young’s double slit experiment. In astronomy it is now essentially a conventional method for obtaining images of objects and structures with very small angular scales (angular scale simply means the angle an object subtends over the sky imagining it as a sphere). The formula for resolution in the telescope is $\Delta = \frac{1.22\lambda}{D}$ hence the larger diameter gives us a smaller more precise resolution for the larger wavelength of radio signals. If we model the light patterns observed in these models, we can predict values for physical properties (just like

fringe separation in Young’s double slit experiment).

The process in astronomy relies on Fourier synthesis, where information from the sky is inversely Fourier transformed into an actual image. Essentially, two or more telescopes observe the same object with a time delay that is precisely measured. These signals from each telescope are then multiplied (convolved) and averaged which produces the fringe pattern known as the complex visibility - essentially a Fourier transform of the sky itself (it is complex as the amplitude encodes the brightness of the signal and phase encodes where on the sky the signal is). As the Earth rotates, these visibility functions can enclose the whole region we want to measure in the sky (u - v plane) which can further be inverse Fourier transformed to reproduce an actual image of the sky. We can sample different spatial frequencies by changing distance between (baselines) and orientation of the telescopes.

After the inverse Fourier transform of the visibility is done, we now have a smeared image of the sky with lots of dirty beams (side lobes and blurry lines - see Figure 1). This is because the VLA has sampled discrete points in space and not the continuous sky itself. To recover the actual sky a process called de-convolution can be applied to the dirty image and there are many algorithms to do this but we will be using

the CLEAN algorithm in this research [11].

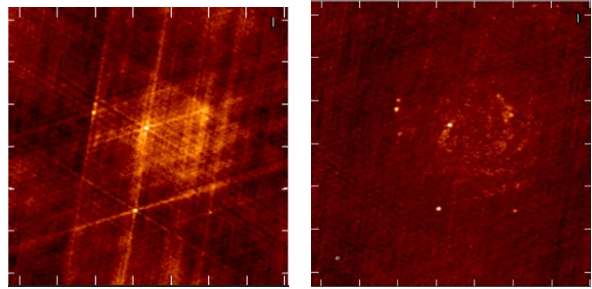


Figure 1: Example image before and after `tclean` has been applied to it, we can clearly see the dirty beams removed. The bright points are real objects (stars or supernovae).

A..4 CLEAN Algorithm

The algorithm starts by generating an interactive image (calibrated images given to us by Deanne from VLA observations) where the real sources (so stars and the supernova) must be specified by drawing boxes over them. The algorithm then finds the brightest point in each of these specified regions (identifies the real source within a specified pixel size, roughly 5-7 pixels that should fit the supernova inside) and then subtracts a scaled dirty beam from this location, while noting the brightness and location of the real sources. This scaled dirty beam is created by multiplying the gain (telescope property which amplifies our original radio signals) with the convolution of the model image (real sources from the brightest pixels) and the dirty beam signals (which were found by specifying the real sources that create the dirty beams/side lobes). We repeat this process

many times (up to 10,000 iterations) until our current image (residual) looks completely like noise meaning we have successfully removed the dirty beams from the discrete sampling of the sky. The algorithm then finishes by re-convolving the model image with the residual to create a cleaned image with our real sources and the noise (before and after shown in Figure 1).

A..5 Measuring the Flux of a Supernova

Once the images are cleaned, the next step is to obtain the actual flux measurements which can then be used to create a radio SED. The process involves drawing a box and obtaining a new image around the location of the supernova in the base image (where the coordinates are easily obtained online from the transient name server). Then to actually run the fit algorithm (imfit from the CASA library), constraints of the supernova must be given (found from the properties of the images given to us by Deanne, including the size of the ellipse we fit and the coordinates etc - more details in method B..2) to the function because the target SN is so far away it is unresolved on the image so the software could easily mistake background noise as the target. The fitting software then fits a 2D elliptical Gaussian to our specified region to extract the flux and its error. Without the con-

straints file, the fit will still work but if a point source is not detected (which would mean a non detection) the measured flux would be significantly higher than the flux measured with the constraints.

One issue that can ruin the fit is galaxy light getting in the way. This can be fixed by varying the weighting (which is a parameter of tclean). In more detail you can change from a Briggs weighting to a uniform weighting which means longer baselines in the data measured by the VLA (consisting of many varying separations between dishes) were favoured and up weighted vs shorter baseline measurements. Longer baselines result in more precise and smaller resolutions meaning small point sources (like the SN) are favoured over large scale structures (like the galaxy). This suppresses the host galaxy's emission while retaining the supernova we target's flux.

A..6 Modelling the Circumstellar Medium

In 2022, De Marchi et al modelled the density structure for SN2004C [2] using methods from [1], a type IIb SN which is very useful to explore since the method is very similar to modelling a type Ic supernova (type II supernovae differ by having more dense and complex CSM environments). 8 years of multi-frequency radio data from

the Very Large Array and Very Long Baseline Array (VLBA) telescope systems was collected with the goal to use the evolution of the radio emission to reconstruct the structure of the CSM and constrain the mass-loss history of progenitor star. They did this by assuming the Synchrotron Self-Absorption (SSA) dominated the epochs of radio data that were observed. The CSM itself actually consists of the gas and dust expelled by the progenitor star before its death through winds and eruptions or interactions with a binary companion. When a supernova blast wave reaches the CSM, the shock wave will accelerate electrons in the magnetic field (which comes from several phenomena; ionized stellar winds, amplification by stellar rotation and shock compression/field amplification by supernova which are explored further in [12] and [13]). These accelerated electrons produce synchrotron radio emission which is what we mean by the SSA radiation. Then each epoch's radio spectral energy distribution was modelled with a broken power law. Multiple parameters were tested to see which fit worked best, then once each epoch was fitted, a break flux and break frequency, properties like the radius of the CSM and magnetic flux density could be computed.

The structure and density of the CSM encode information about the progenitor star's final phase of life. Examples include

smooth density profiles which imply steady winds, flat or broken profiles (observed in SN2004C [2]) which instead imply variable mass loss possibly due to binary interactions.

Moving back to type Ic SN (specifically our type Ic SN, 2017ein), the same process is essentially undergone which we will explore more specifically.

A..7 The Model We Will Use for SN2017ein

Repeating the methods used in [2] which were also used in [1]), we will specifically show how we model the CSM of SN2017ein.

The parameters we want to calculate are magnetic flux density (B), radius (R), mass loss rate ($\dot{M}$) and density (ρ). The post supernova shockwave sent out by the explosion when it started has a forward and reverse shock which would be separated by a discontinuity. The radio emission which we observe is dominated by synchrotron emission from the accelerated electrons at the forward shock wave which can be modelled with a power law distribution [1]:

$$\frac{dN_e}{d\gamma_e} = K_0 \gamma_e^{-p}, \quad \text{for } \gamma_e \geq \gamma_m \quad (1)$$

,

where γ_m is the minimum Lorentz factor of all the electrons, γ_e is the electron Lorentz factor, N is the number of electrons

per unit volume, K_0 is the normalisation constant and p is the power law index for the electron distribution. Standard radio observations indicate p is approximately 3 for typical SN electrons which gives a very steep distribution so we can assume the low energy electrons dominate the curve and very few electrons will have γ above γ_m due to the sharp cut off. This means we can assume that $\gamma_e \approx \gamma_m$.

We then need to set up how the energy is split up, which is split between the magnetic field and the electric field with microphysical shock parameters ϵ_B and ϵ_e . The energy density is $\rho_{CSM}v_{shock}^2$ [2], and

$$E_B = \frac{B^2}{8\pi} = \epsilon_B \rho_{CSM} v_{shock}^2 \quad (2)$$

$$E_e = \epsilon_e \rho_{CSM} v_{shock}^2 \quad (3)$$

where ρ_{CSM} is the CSM density and v_{shock} is the blast wave's forward moving velocity. There are two sets of values of ϵ we can adopt. The first are the fiducial values which can be approximated as $\epsilon_B \approx 0.01$ and $\epsilon_e \approx 0.1$ [2]. These can be thought of as the more physical interpretation of the values and then we also will calculate the equipartition values where $\epsilon_B = \epsilon_e = 0.33$ and these are interpreted as more ideal values which are only useful for comparing radio modelled SNe (which is what we will do). Note the extra 0.33 we are ignoring come from the proton energies which are

not relevant this research as proton radiation is not detected. Some sources used in our discussion will use different values for equipartition but once our final equations are derived we will see scaling for different equipartitions has a simple solution.

[14] shows how when we assume all our electrons have the minimum γ value from Equation 1, we have a direct dependence of the peak frequency observed on B [2]

$$\nu_m \approx \epsilon_e^2 \left(\frac{v_{shock}}{c} \right)^4 \left(\frac{m_p}{m_e} \right)^2 \left(\frac{eB}{2\pi m_e c} \right) \quad (4)$$

where m_p and m_e are the respective masses of the proton and electron, c is the speed of light and e is the electric charge. We now have a break frequency which divides the overall radio spectrum (measure of brightness) into optically thick and thin regions (forwards shock creates the thick region and reverse shock does the opposite). In [2], a large equation for specific intensity which uses equation 4 (I_ν) is derived and parameters of this equation are defined in [15]. In these equations the radio emitting region is assumed to be a disk in the sky with radius R (one of the desired physical parameters). With the specific intensity we can now link the flux to the frequency by using the equation $F_\nu = I_\nu \pi R^2 / D^2$ [2] where D is the distance to the radio emitting disc. We can then split up into a thick and thin region flux by assuming the fre-

quency is a lot larger than the break frequency for the thick region and the opposite for the thin region. Without going into detail yet on what the equations look like we are now at a point where we have a flux for a thin and thick region depending on frequency, B and the radial distance from the centre of the explosion (where the supernova first occurred). We can equate the thin and thick flux densities at a certain frequency which we will now refer to as the break frequency (ν_{brk}) and then call the flux density correlating to ν_{brk} the break flux (F_{brk}). To finally reach physical equations for B and R we must understand how B depends on the power law index for electrons ($p \approx 3$). $E_e/\epsilon_e = E_B/\epsilon_B = \rho_{CSM}v^2$ and using this result along with equation 3 and equation 2 and the lowest electron energy ($E_l = \gamma_m m_e c^2$) [2] (which comes from our earlier electron power law equation 1 - introducing p), we can derive a relation between B and p which is done in detail in DeMarchi et al 2022 [2] allowing us to finally form equations for B and R which depend on F_{brk} and ν_{brk} . To simplify the equations we will state them evaluated for typical radio SNe values ($p = 3$ and $E_l = m_e c^2$) [2] reducing our B and R to:

$$B = 0.5761 \left(\frac{\nu_{brk}}{5 \text{ GHz}} \right) \left[\epsilon_B^{-1} \left(\frac{D}{\text{Mpc}} \right) \epsilon_e \left(\frac{F_{brk}}{\text{Jy}} \right)^{1/2} \right]^{-4/19} \text{ G} \quad (5)$$

$$R = 8.759 \times 10^{15} \left(\frac{\nu_{brk}}{5 \text{ GHz}} \right)^{-1} \left[\left(\frac{D}{\text{Mpc}} \right)^{18} \epsilon_B \left(\frac{F_{brk}}{\text{Jy}} \right)^9 \epsilon_e^{-1} \right]^{1/19} \text{ cm} \quad (6)$$

Where distance is in Mpc (a parsec is defined as the distance to an object that subtends an angle of $1/3600$ with the average radial orbit of Earth around the Sun) and the flux is in Jy (Jansky - defined as $10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}$).

We also stated we want to find the mass loss rate and the density (which are dependent on each other) simply linked by the equation [2]:

$$\rho_{CSM} = \frac{\dot{M}}{4\pi R^2 v_w} \quad (7)$$

where $\dot{M}$ is the mass loss rate of the supernova and v_w is the velocity of the wind (outward radiation "pushed" from the explosion). This equation only holds valid if we assume that the winds of the supernova are modelled as steady and spherical.

All that is left now is to derive how we model the mass loss rate. The full derivation is beyond the standard of this report so we will state the final equation but a simplified explanation follows. By combining equation 2 with equation 5 and then assuming a power law for R ($R \propto t^q$ where q is the power law index), and then applying

equation 7 we reach the equation [2]:

$$\begin{aligned} & \frac{\dot{M}}{v_w} \left(\frac{1000 \text{ km s}^{-1}}{10^{-4} M_{\odot} \text{ yr}^{-1}} \right) = \\ & 2.5 \times 10^{-5} \left(\frac{\nu_{\text{brk}}}{5 \text{ GHz}} \right)^2 \left(\frac{t}{\text{days}} \right)^2 \epsilon_B^{-1} \\ & \times \left[\epsilon_B^2 \left(\frac{D}{\text{Mpc}} \right)^{-2} \left(\frac{F_{\text{brk}}}{\text{Jy}} \right)^{-1} \epsilon_e^{-2} \right]^{4/19} \end{aligned} \quad (8)$$

,

which has been evaluated for $q = 7/8$ corresponding to a CSM created by massive stellar winds and a core collapse supernova originating from a WR star [16]. We also note that to convert to physical units we must multiply by the physical conversion constant $1000 \text{ km s}^{-1} / 10^{-4} M_{\odot} \text{ yr}^{-1}$. This concludes the modelling used to find our values but one final point to take into consideration is to compare to values derived based on different equipartitions, you must scale. For R and B the scalings cancel out but for $\dot{M}$ and similarly ρ_{CSM} we must scale by $\epsilon_{\text{lit}} / \epsilon_{\text{ours}}$ where ϵ_{lit} is the literature value for equipartition and ϵ_{ours} is our value for equipartition which as stated before is 0.33.

A..8 SED fitting

Now that there is a framework for the modelling, we need to understand where our physical data fits into the modelling. From the imfit measurement, flux density val-

ues are extracted (in mJy) and then meta data on each image reveals the frequency. So from our modelling we must extract an equation which links the flux to the break flux, break frequency and the frequency. Recall the definition of the break flux is when the thick and thin regions meet (at the break frequency). From [14] the thick region flux equation scales with $F_{\nu, \text{thick}} \propto \nu^{5/2}$ and the thin region flux equation scales with $F_{\nu, \text{thin}} \propto \nu^{-(p-1)/2}$. This can then be combined to create a broken power law (sum of individual parameters scaling with a certain power):

$$F_{\nu} = F_{\text{brk}} \left[\left(\frac{\nu}{\nu_{\text{brk}}} \right)^{\alpha_1/s} + \left(\frac{\nu}{\nu_{\text{brk}}} \right)^{\alpha_2/s} \right]^s \quad (9)$$

,

where we already know $\alpha_1 = 5/2$ in our modelling so that can be constrained in the fit as a theoretical assumption. $\alpha_2 = -(p-1)/2$ and s is known as the smoothing parameter which represents how abruptly the SED turns over from optically thick to optically thin. This can be constrained to a value of -1 as done in [2]. Then observationally we can expect the range of α_2 to be in between 0 and -3 as a final constraint [2, 16].

Using scipy in python, data from imfit can be plotted as flux against frequency (being careful about units) and then the

function `curve_fit` can have equation 9 input into it to try fit the equation to the plotted data and then predict the values for F_{brk} , ν_{brk} and α_2 .

B. Method

B.1 Cleaning Images

We implemented this algorithm using the function `tclean` provided by the python library CASA in a linux environment. We had received one image for the first epoch and then five for the second epoch (third epoch has not been used in this project due to the time we had). Each image corresponded to a certain wavelength band (L - 1.4 GHz, S - 3 GHz, C - 6 GHz, X - 10 GHz and K - 22 GHz) and when running the `tclean` algorithm we had to input the number of spectral windows (essentially range or broadband of frequency - higher frequency means more spectral windows), select a cell size such that 5-7 pixels would fit inside a reference shape on the interactive image (from the algorithm `imview` in CASA) and other standard parameters. After cleaning each image, to confirm it was done correctly, the residue of the image was observed to see if it consists of only noise.

B.2 Flux Measurement of Supernova

To run the `imfit` function, a few parameters to give the function had to be found first.

We made a target region file to give the algorithm so the background emission could be seen, and then we gave the constraints on the supernova (major axis of the point source ellipse, minor axis, angle, flux and pixel values of supernova). We also measured the background values median RMS (root mean square) value by creating several boxes and extracting the median RMS values from these boxes (around the supernova). This was to check if we have a non detection which we defined using a 3-sigma threshold. If our value was under 3 times the median background RMS value then it would be defined as a non-detection and not be used for the SED fit (the fit on a plot of fluxes against frequency for the radio data).

After we set up the constraints and had a region file and median RMS value measured, we were able to run the `imfit` function to extract a flux value and its error. The value and error is found by fitting a 2D elliptical Gaussian model to the constrained location of the supernova. The fit also created a residual file and if we observed a hole where the supernova should have been it means the background galaxy light also contributed to our measurement and that the weighting needed to be changed to uniform as explained in the theory A..5.

B..3 Spectral Energy Distribution Fitting

Once we had a list of flux measurements and their errors we were able to simply plot this as a scatter plot in python (using matplotlib in a Jupyter notebook environment) of flux density with errors against the corresponding frequency for each flux which made up our SED.

To get our break flux and frequency values we used the scipy library and then imported the function `curve_fit` and wrote equation 9 as a function to input into the parameters for the fitting algorithm. We also set the errors as our weightings to ensure accurate errors are found and then applied the same constraints mentioned in the theory A..8. The program then used the least squares method to fit a curve to our points and predict break values and alpha 2. The errors themselves were found from by the algorithm finding the diagonal values of the covariance matrix created from our errors.

To help boost the fit's accuracy, it was also possible to split our images into two measurements by splitting each frequencies broad band up into two (essentially each filter into two sub filters where mid points are used for frequency values) and then repeating the clean and flux extraction process which would give you more data points to fit for.

B..4 Calculating Parameters to Model CSM

After finding fitted break flux and frequency values along with their errors, all that is left to find to model the physical parameters is the distance to 2017ein and the time since explosion. We could simply find these values looking at past literature ([6] for the time since explosion as a date then take the difference of that with the time of our data measurements, again stated as a date in the image properties) and [9] had measurements of distance published (in distance magnitude which was converted to Mpc). Once we have the distance, time since explosion, break flux and break frequency for our values equations 5, 6, 7 and 8 could be used to find the physical parameters for magnetic field, radius, mass loss rate and density.

To find the errors on our physically modelled parameters, we assumed that the errors from the researched values (distance and time) and our fitted values (break flux, break frequency) were all independent so standard error propagation could be used to find analytical solutions for the errors using partial differentiation.

For distance and time, we could simply find the errors for the distance magnitude and time of explosion as they were stated in [6, 9] and simple unit conversions gave us the error for these values. (Time

was just a difference with the time of our observations so the error did not change and the distance magnitude conversion formula is $d(Mpc) = 10^{(u-25)/5}$ [17] (converted to Mpc here) where u is the distance in magnitude so the error can be found from $\sigma_d \approx 0.4605d\sigma_u$ using the standard error propagation formula.

B..5 Collecting Data for Similar Supernovae Comparison

To conduct the comparison study for 2017ein, we had to research and find values from previously published literature. We tried to find similar properties (B , R , ρ_{CSM} and $\dot{M}$ for different type Ic, type Ib (similar modelling) and radio type II SNe.

If any researched values were stated as a range in previous studies, we found errors by simply calculating half of the range and set the value as the midpoint of the range and the error as plus or minus half the range so our value spans the full range given. Many considerations and assumptions were used to collect these values which will be explored in our results.

III. Results

A. Flux Measurement Results

After applying the CLEAN algorithm on each image in the epoch that we used (epoch 2 of images received from Deanne),

imfit was applied on each image to extract the flux and error required. The frequency is already known from images taken being in specific wavelength bands in the radio range.

Band	Frequency (GHz)	Flux (mJy)	Error (mJy)
C	4.999	0.34828	0.07440
C	7.099	0.30676	0.08804
L	1.255	2.24038	0.68264
L	1.739	0.86709	0.13473
X	8.999	0.18573	0.04086
X	10.999	0.16745	0.06541
S	2.599	0.74694	0.00739
S	3.399	0.60158	0.00660
K	22.064	0.24838	0.23277

Table 1: Results of applying imfit to each image in epoch two. Note that each band except K was split into two images halfway across band broadband. K was not due to being under the 3-sigma threshold as explained in the method. Also note our lower frequency L band measurement is a non detection to due failure of a point source being detected.

Table 1 shows the values of flux and their corresponding errors extracted from the fitting process. Note there are some non detections which will be shown on the SED fit and will be explored soon.

In the theory for flux measurements A..5 it was mentioned that the fit may not work if too much galaxy light is measured along with the supernova. When we first applied the imfit algorithm to our cleaned images, we found every image’s residual had a dark hole where the galaxy light (including the supernova) would have been implying that the galaxy light was getting in the way and our algorithm was trying to fit the galaxy itself. This resulted in our fits repeatedly failing or giving erroneously large values. To fix this, we changed from the Briggs

weighting to the Uniform weighting favouring the larger baseline wavelengths as explained in the theory A..5.

B. Spectral Energy Distribution Fit Results

Now that we had measured the flux and the error from our imfit processes we then plotted the flux measurements against the corresponding frequency values. We then applied the fit from equation 9 to extract the values for the break flux, break frequency and α_2 . The fit was constrained using the justifications in the theory for SED fits A..8

. For the fit seen in Figure 2, the data was split into two points for each wavelength band. When we initially tried to fit the SSA model (equation 9), the fit was unable to identify accurate parameters without reaching infinities due to the fact that we had essentially 4 data points to fit a model for 3 parameters which in general is not enough to constrain the fit. This then doubled the number of data points to 8 (in the end we had 7 for the fit as not all the points turned out to be useful). 8 data points for a 3 parameter fitting model was still not enough without constraints. As explained in the theory (A..8) there were several possible constraints that could be applied and one of the constraints involved setting a range for the parameter

α_2 and this was also possible to do so for all the required fit parameters within physical meaning but this method of constraining could also easily "force" a fit on our SED. We had initially tried to constrain the break flux and frequency within the range of our values measured from the previous imfit method. While the constraint on the range of flux ($[0, 5]$ mJy) turned out not to affect the model (which is a justifiable range as it is very broad and large), our first range on frequency $[1.255, 10.999]$ GHz (the range of our data points) repeatedly forced our fitted flux on 1.255 GHz itself (our lower L band value) and then further loosening the constraint to start from 0 GHz still would force our fit at the first L band value. The only way to fix the fitting was to set the lower L band as a non detection which turned out to be the case as we repeated the imfit flux measurement but found that it was not a point source being found which means a non detection (A..5).

The fit can be seen in figure 2. We can see two non detections, one for the lower frequency L band measurement where a point source fit was not found and then one for the K band value where the flux was found to be below our selected 3-sigma threshold.

The fitted flux found was $F_{brk} = 1.54 \pm 0.07$ mJy and fitted frequency was $\nu_{brk} = 1.6 \pm 0.2$ GHz. We also found a value of

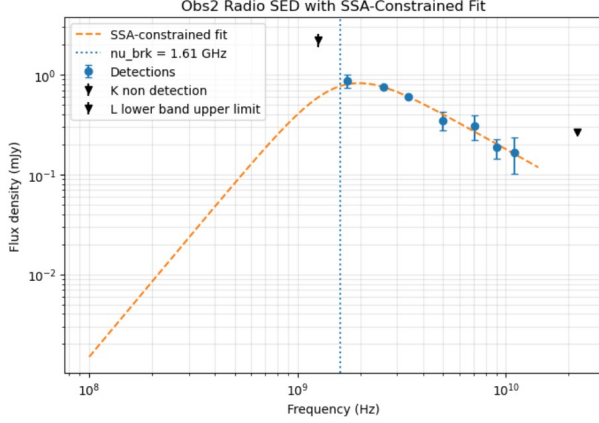


Figure 2: SED fit for our flux measurements from the radio images. The black arrows correspond to non detections and the orange line shows the model fitted using CITE EQN HERE TOO

$\alpha_2 = -1.2 \pm 0.2$ which then using $\alpha_1 = -(p-1)/2$ [2] we infer a value of p to be $p = 3.3 \pm 0.3$ in agreement with the physical value $p \approx 3$ [16].

C. Modelling Parameters

Using equations 5, 6, 8, $t_{days} = 12 \pm 0.3$ days (found from the difference in the date of our data measurement and the date when explosion occurred [6]), $D_{Mpc} = 19.1 \pm 3.5$ Mpc (found from [9] which stated $D = 31.4 \pm 0.4$ mag which was converted from magnitude to Mpc) and our fitted values, we find for equipartition ($\epsilon_B = \epsilon_e = 0.33$), $B = 0.20 \pm 0.03$ G, $R = 2.1 \pm 0.5 \times 10^{16}$ cm, $\dot{M}/v_w = 9.4 \pm 3.0 \times 10^{10}$ g/cm and then using equation 7, we find $\rho_{CSM} = 1.7 \pm 1.0 \times 10^{-23}$ g/cm³. For fiducial ($\epsilon_B = 0.01, \epsilon_e = 0.1$) we find $B = 0.12 \pm 0.02$ G, $R = 1.8 \pm 0.4 \times 10^{16}$ cm, $\dot{M}/v_w = 1.2 \pm 0.4 \times 10^{12}$ g/cm and $\rho_{CSM} = 2.7 \pm 1.6 \times 10^{-22}$

g/cm³.

D. Comparison to Supernova Sample

Now that we have physical parameters for SN2017ein's CSM, we can compare this to other radio modelling of similar supernovae (e.g. compare densities). Table 2 shows the same parameters for different supernovae (along with an extra parameter useful for comparison - shock velocity which will be discussed in more detail soon). We will go into the considerations taken into account to collect this data soon but first we will present plots of the variations of the physical properties compared for different supernovae. Note that all of the values have been scaled to our equipartition ($\epsilon_B = \epsilon_e = 0.33$) from equipartition values used in other derivations with the same model.

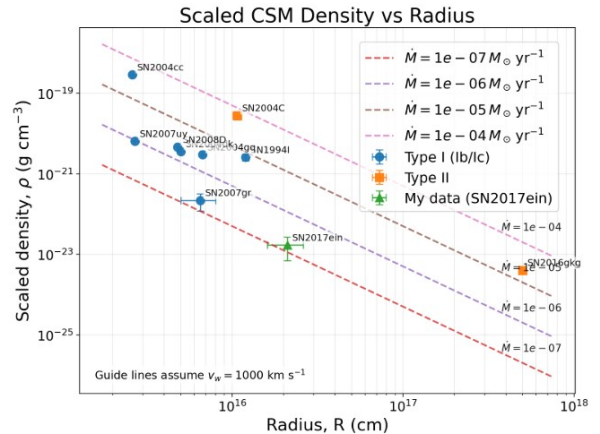


Figure 3: Density scaled to our equipartition vs radius/radial distance from the initial explosion for each supernova identified in Table 1 (including our data too). Type I and Type II are split up. Scaling lines show how different densities fit on different mass loss rates.

Figures 3 to 6 show how our physical results compare to the sample of similar supernovae found from other studies [18, 19, 20, 21, 22, 23, 24] listed in Table 2.

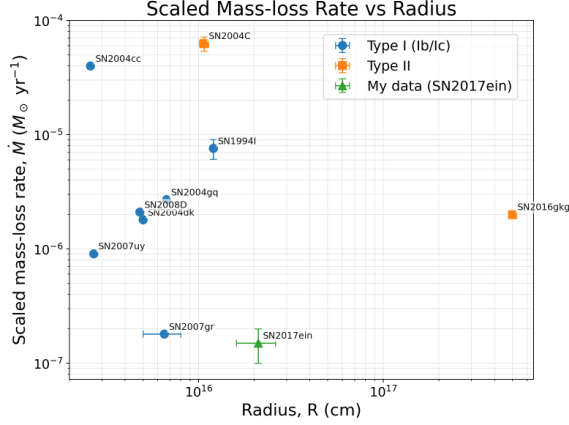


Figure 4: Mass loss rate scaled to our equipartition vs radius for the supernova sample. Mass loss rate found from (8) then converted into units of solar mass per year.

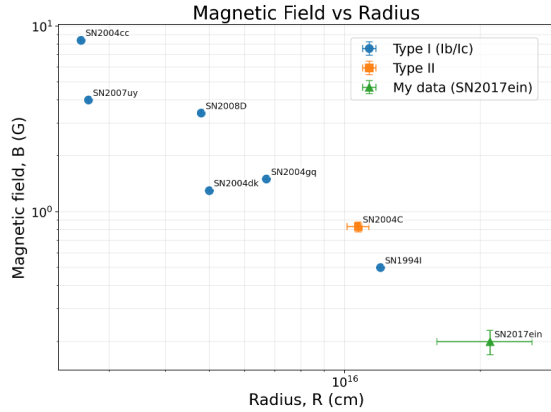


Figure 5: Magnetic field strength vs radius for the supernova sample. Magnetic field found from (5).

The plots shown compare scaled values of density vs radius, scaled values of mass loss rate vs radius, the magnetic flux density vs radius and then the calculated shock velocity vs an arbitrary x axis. Note that when we state radius we actually mean physically

the radial distance from where the explosion initially occurred. These plots can be used to compare properties of SN2017ein to similar supernovae which will soon be explored in the discussion section.

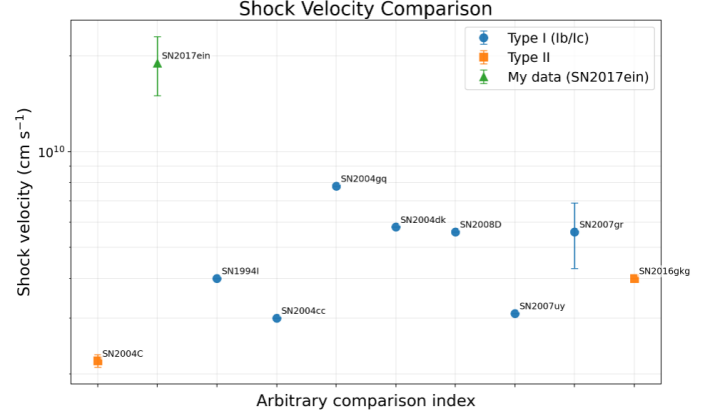


Figure 6: Shock velocity (simplified speed of forward shock wave) comparison across the supernova sample. An arbitrary x axis was used here which was simply the index of the supernova. Note in all our plots the green point is corresponding to our data point.

Certain points and assumptions must be taken into consideration when looking at the comparison results.

Physically, the shock wave of the supernova is moving in real time, hence physical parameters like radius will also vary in time. Previous radio modelling studies may have several results for different epochs (for example table 5 in [2]) after the explosion of SN2004C so it was very important that when we use a physically modelled radius, we make sure we use the other physical parameters corresponding to the same epoch in time for said supernova due to the fact we are plotting parameters against radius. The issue remains that if we have a cer-

tain epoch for physical values, in a previous epoch for example the density could be significantly lower due to something like eruptive mass loss and we will not have any record of that.

Initially the plan was to use wind velocity values to compare different supernovae but after conducting the research it was understood that wind velocity is a loosely constrained value and is approximately equal ($v_w \approx 1000$ km/s) for many type Ib/Ic supernovae which share having a Wolf-Rayet star for their progenitor [25]. It then made more sense to calculate and compare a simplified value of shock velocity, which can be calculated assuming radius and time since explosion for when this radius is measured were known from the formula $v_{shock} = R/T$. This result must be treated with the strongest caution as the results will be extremely model dependent (figure 6) as it is a very simplified equation being used.

In the theory for CSM modelling (A..6) it was mentioned we must assume steady spherical winds to find the density from the mass loss rate using the simple equation 7. Many of the sources stated the mass loss rate which then we had to assume the steady spherical wind to calculate the density ourselves. For the types of supernovae we see here (radio SNe) this is a valid assumption to make. [26] establishes the r^{-2} wind profile used for radio models

and for supernovae with WR star progenitors (most type Ib/Ic), the winds are fast and relatively steady hence the assumption holds out to find our density values.

From the comparison plots you may have noticed there are not many radio type II supernovae we were able to find physically modelled parameters for. Type II supernovae commonly arise from red supergiant stars which tend to have slower winds by at least a magnitude in comparison to WR stars hence we should expect them to have denser CSM environments at a given mass loss rate. This leads to a more complicated radio spectrum with episodic and asymmetric mass loss straying away from the r^{-2} proportionality because of a larger affect from free-free absorption from the CSM itself (free-free absorption occurs when "free" ionised electrons absorb a radio photon and remain free removing the radio photon from observation and if the CSM is more dense this is more likely to happen). Hence overall it is harder to physically model parameters for type II supernovae.

In the CSM theory, A..6 it was stated that for different equipartition values you can simply scale by a factor of the ratio of another equipartition value to our equipartition value (0.33). Most the other literature used equipartition values of $\epsilon_B = \epsilon_e = 0.1$ so we had to correct mass loss and density values by a factor of $0.1/0.33$. This is

acceptable to do when only looking at other radio modelled parameters but the issue is if there are cases where there are errors on the originally published values these errors get scaled which may not be completely physically accurate.

You may also notice in Table 2 that not all the values have errors. This is simply due to the fact that not all the papers we researched published errors in the values. This is because as seen in A..6 there are many assumptions that need to be made to reach our equations for the physical parameters hence it is a standard practice in radio astronomy that it is not necessary to to consider errors that would be very small relative to the errors caused by the assumptions made in the first place. Taking these considerations into account we can move on to actually explaining what the results and comparisons teach us about SN2017ein.

IV. Discussion

Before looking at the final model parameters and what they say about SN2017ein, it is essential that we keep in mind all the assumptions that were taken into account to build the comparison table (Table 2).

A. How Does SN2017ein Compare in Population Study

We will now explore what our comparison plots actually tell us about 2017ein which

will lead us into what we can learn about its progenitor.

A..1 Density of SN2017ein Compared To Other Radio SNe

Starting with the density plot we can very clearly see that half of the type Ib/Ic supernovae sit together (between the lines corresponding to 10^{-6} to $10^{-5} M_{\odot}/yr$) we observe SN2017ein having a very low relative CSM density compared to the half (5) SNe we see grouped together between 10^{-6} and 10^{-5} . We can also see the the type II supernovae that we have sit on a straight line corresponding $10^{-4} M_{\odot}/yr$ implying that type II radio SNe tend to have more dense complex CSM environments, but since there are only two data points more type II supernovae would be required in the sample to support this claim solely on our data but recall we did mention earlier in the supernova theory (A..1) type II supernovae have larger densities and more complicated CSM environments which is supported by this plot. Recall the equation used for density, 7, for given mass loss rates from this equation, we expect a log linear dependency on radius and this is observed for all of the type I SNe in our plot (any points off the line do not go over a 3-sigma value). Vaguely speaking this implies overall higher radius/distance to origin of explosion means lower density. 2017ein fol-

lows that trend but the density is lower than expected for a type Ic SN. [16] states typical radio modelled type Ib and Ic densities lie within 10^{-24} to 10^{-21} g/cm^3 , this would mean our density for SN2017ein lies at the lower end of the densities having a magnitude 10^{-23} . A low density would imply a relatively clean CSM environment and weaker mass loss rates or less strong winds before the explosion. This would agree with [10] which claimed 2017ein had a low ejecta mass. Our derived radius is also relatively large at a time of 12 days post shock implying a large shock velocity (this is observed in Figure 6). This is also consistent with the fact that a low density environment would not be able to significantly affect the forward shock wave hence a large shock velocity. In terms of what we can learn about the progenitor, the density level sits in the typical range for WR stars but the claim that 2017ein also had low ejecta mass would imply a binary interaction for the stripping mechanism. As explored in the methods (III.D.) there are still many reasons that 2017ein could be an outlier. Firstly we are looking at different epochs for each supernova which can affect the results, also while the equipartition values were scaled even the slightest errors can vastly affect the physical values and as mentioned our SED had sparse data for the fit. Overall we can conclude that there is a strong amount of evidence implying

SN2017ein had a sparse CSM environment with relatively weak winds pre-explosion but also there is not strong enough data to conclude solely based on the density of what the progenitor is but from [10], the claim of a stripped binary progenitor is supported. Also note this plot shows SN2004cc being very dense too which is implied in [19] due to very strong radio emission unlike SN2017ein.

A..2 Mass Loss Rate Of SN2017ein Compared To Other Radio SNe

Since we assumed steady spherical winds, the mass loss rate and density are closely related so already we expect a relatively low mass loss rate especially due to the fact we implied weaker mass loss rates pre-explosion from Figure 3. This is exactly what is observed in Figure 4 and again we see a similar trend because half of the supernovae are grouped together but again SN2004cc is an outlier on the upper end and SN2017ein (and SN2007gr) are outliers on the lower end (all by over a magnitude, further implying the weak mass loss rate pre-explosion). The mass loss rate essentially tells us "how heavily the progenitor loaded its surrounding environment with wind material" [1], and the relatively low value ($1.5 \pm 0.5 \times 10^{-7} M_{\odot}/\text{yr}$) sits outside the claim that classical WR stars have

strong radiative driven winds within the range of 10^{-6} to $10^{-5} M_{\odot}/yr$ [27] (exactly what we see on Figure 3. Against this background SN2017ein is further supported to come from a binary stripped scenario versus a singular classical WR star as this mass loss rate sits comfortably within the range of binary stripping mass loss rates [5]. While the mass loss rate plot is consistent with the density plot and a binary stripped scenario is implied, we again need to take into consideration that many assumptions have been applied to produce this plot so we cannot fully constrain the progenitor as a binary stripped star. Overall this plot implies that single classical WR stars are not the only type of progenitor for type Ib/Ic radio SNe but further research is required to confirm a binary stripped scenario.

A..3 Magnetic Field Of SN2017ein Compared To Other Radio SNe

Now moving onto Figure 5, we see a more consistent plot overall with a clear relation showing that as radius or the distance from the explosion increases the magnetic field decreases. While the magnetic field alone cannot tell us much about the progenitor it still remains consistent with our discussions so far. From the CSM theory A..6 we stated that the radiation we detect is caused by synchrotron self absorp-

tion emission. B is linked to the density and shock velocity with scaling $\propto \sqrt{\rho}v_{shock}$, [1]. On Figure 5, SN2017ein has the lowest magnetic field value even though this is expected further away from the explosion. The typical range for radio modelled type Ib/Ic radio SNe lies within 0.3 to 3 G [21] so our value (0.20 ± 0.03 G) is still relatively low. This can be explained due to our relatively very low density having stronger scaling even compared to the shock velocity weighting. Physically speaking the magnetic field strength relates directly to the energy of the shock wave and since it is already established that SN2017ein has weak radio luminosity and low density a low magnetic field is also expected. Note also we see more energetic supernovae like SN2004cc with a considerably larger B value implying the correlation between energy/density and the magnetic field. Overall we can conclude that our value for B is self-consistent with our SSA modelling but again it is highly model dependent due to scaling of epsilon values and in general ϵ_B and ϵ_e not being well constrained observationally.

A..4 Shock Velocity Of SN2017ein Compared To Other Radio SNe

The shock velocity calculated will need the most precaution as we have calculated this

value by simply dividing our SSA-derived radius by the time since the explosion. While the time has been found from a credible source ([6]), the radius has been found from our relatively sparse data set. This precaution is shared by the rest of the plots but Figure 6 needs the most care as the actual equation used assumes a simple smooth blast wave with constant velocity which is not physically accurate. However, we can confidently state that 2017ein once again looks like an outlier (the most large in this case), the value itself is roughly $0.6c$ - physically unlikely as this would indicate some extremely exotic engine driven event with one of the most rapid and extreme blast waves ever detected. But looking at the density, physically a low density value would mean the shock wave moving forward would not have much matter to oppose it hence a relatively fast shockwave would be expected. The rest of the supernovae appear to have velocities closer together with the type II supernova SN2004C having the lowest value (supporting the earlier claim of type II being more dense). Overall, we can confidently say our large shock velocity value is consistent with the other plots but the value itself is not physical due to over simplifying assumptions.

B. What can we learn about the Progenitor of SN2017ein

Combining what has been inferred about each plot separately, our data shows us that we have a very simple CSM with slow steady mass loss before the explosion. The relatively low values of density, mass loss rate and magnetic field are all self consistent then a high shock speed is a physical consequence of these properties. The mass loss rate gives the strongest indication out of all the data that the progenitor of SN2017ein is unlikely to be a classical large WR star and more likely it was stripped from a binary mechanism. While the data is self-consistent, due to the large amount of significant assumptions made in the modelling and data collection discussed in the methods (solely relying on one epoch for each supernovae values, assumed the less physical more ideal ϵ values, simple density and shock velocity equations based on assumption of steady winds, etc), we cannot claim that this confirms a binary stripped progenitor but rather implies it. We also must recall we had a small amount of data initially for the fit to calculate our modelled values and even classical WR progenitors can still have mass loss rates and densities lower than the quoted range of 10^{-6} to $10^{-5} M_{\odot}/yr$. Recall also in the theory about SN2017ein (A..2), recent studies state the supernova is a low luminosity less

dense supernova [5] which is aligned with the direction our data leads us in. This is interesting because our results show us that type Ic supernovae may not only come from single massive WR stars but that it is likely a lower mass WR star with a binary companion can also cause these certain types of supernovae.

C. What Could Be Done Differently?

Before we conclude the report, it is good practice explore what could be improved if this research was to be reproduced. It is worth noting that we had 2 terms (roughly 20 weeks) to conduct our research and some of these improvements would likely need more time to fully be undertaken. Lets start with the number of data points we had. In the results for the SED (III.B.) we stated that we had 7 data points to fit for three parameters. It would have vastly improved our accuracy if we had used more data points (a third epoch of data was available but due to time constraints we did not get to analyse these). Along with the sparse amount of data we also had a few simplifying assumptions which would have benefited to have more thought into them. Starting with the spherical winds assumption leading to Equation 7, When we assumed the steady spherical winds, further investigations could have gone into each

separate supernova to see if the assumption fully applies or if the density calculation needs any sort of corrections for more energetic supernovae. Another assumption we have taken for granted is the wind velocity being roughly 1000 km/s for the type Ib/Ic supernova. The wind velocity was used to calculate density and mass loss rates so a lot of our data does rely on it and different values could have taken this project in a different direction. In general it is a poorly constrained value and we assumed the single WR star progenitor values when our results imply a binary stripped progenitor. Again further investigation could have gone into a more specific wind velocity value to use although this is subjective as the value is not well constrained in the first place. Along with assumptions made in our methods, the data collected for the table may not all be extremely accurate (especially simplified shock velocity calculations) as different scalings and assumptions were used to even create the table. In general, in radio astronomy there is a large cost to take images and to get several epochs of data it is not necessarily guaranteed, so the earlier point we made about only having one epoch for our data is a broader improvement that would need more resources rather than simply more time. One final point we make is that it was already known that SN2017ein was a faint source so naturally some images would

be harder to analyse and accurately extract fluxes from (hence we observed two non-detections). Overall, some of these improvements could have definitely been made if more time was available but some improvements also solely rely on past observation and data available (likely to improve as time goes on and more observations are taken).

V. Conclusion

We now come to end of our attempt to model the circumstellar medium of SN2017ein. Overall the results have implied to us that we are dealing with a low density radio supernova with a relatively clean and sparse environment around it. The key results of $\rho_{CSM} = 1.7 \pm 1.0 \times 10^{-23} \text{ g/cm}^3$ and $\dot{M}/v_w = 9.4 \pm 3.0 \times 10^{10} \text{ g/cm}$ imply the supernova came from a binary stripped progenitor rather than a single classical massive Wolf-Rayet star when we compare to other similarly modelled radio supernovae. But due to time constraints, data availability and simplification we cannot claim this is the official case, we can only conclude our data is self consistent (with other physically modelled values $B = 0.20 \pm 0.03 \text{ G}$ and $R = 2.1 \pm 0.5 \times 10^{16} \text{ cm}$) to reach these conclusions. To understand why we modelled what we did we also explored deeply the theory behind radio astronomy and SSA derived radio modelling

leading us to perform a broken power law fit on a spectral energy distribution which we created from radio images received by our supervisor Deanne. We also stated ways the research could be improved if further similar research was to be conducted into SN2017ein.

Supernova	Type	B [G]	Radius [cm]	Shock velocity [cm s ⁻¹]	Density (scaled) [g cm ⁻³]	$\dot{M}_{\text{scaled}}$ [$M_{\odot}$ yr ⁻¹]	Reference
SN 2004C	IIf	0.83 ± 0.05	$(1.07 \pm 0.06) \times 10^{16}$	$(2.2 \pm 0.1) \times 10^9$	$(2.8 \pm 0.6) \times 10^{-20}$	$(6.3 \pm 0.9) \times 10^{-5}$	[2]
SN 2017ein	Ic	0.20 ± 0.03	$(2.1 \pm 0.5) \times 10^{16}$	$(1.9 \pm 0.4) \times 10^{10}$	$(1.7 \pm 1.0) \times 10^{-23}$	$(1.5 \pm 0.5) \times 10^{-7}$	—
SN 1994I	Ic	0.5	1.2×10^{16}	4.0×10^9	$(2.6 \pm 0.5) \times 10^{-21}$	$(7.6 \pm 1.5) \times 10^{-6}$	[18]
SN 2004cc	Ic	8.4	2.6×10^{15}	3.0×10^9	2.92×10^{-19}	4.0×10^{-5}	[19]
SN 2004gq	Ib	1.5	6.7×10^{15}	7.8×10^9	3.0×10^{-21}	2.7×10^{-6}	[19]
SN 2004dk	Ib	1.3	5.0×10^{15}	5.8×10^9	3.6×10^{-21}	1.8×10^{-6}	[19]
SN 2008D	Ib	3.4	4.8×10^{15}	5.6×10^9	4.6×10^{-21}	2.1×10^{-6}	[20]
SN 2007uy	Ib	4.0	2.7×10^{15}	3.1×10^9	6.5×10^{-21}	9.1×10^{-7}	[20]
SN 2007gr	Ic	—	$(6.5 \pm 1.5) \times 10^{15}$	$(5.6 \pm 1.3) \times 10^9$	$(2.2 \pm 1.0) \times 10^{-22}$	1.8×10^{-7}	[21]
SN 2020oi	Ic	—	$(2.0 \pm 1.0) \times 10^{15}$	—	$(1.2 \pm 1.2) \times 10^{-19}$	1.0×10^{-5}	[24, 28]
SN 2016gkg	IIf	—	5×10^{17}	4.0×10^9	4.01×10^{-24}	2.0×10^{-6}	[22]
SN 2004dj	IIP	—	$(3.0 \pm 1.0) \times 10^{15}$	—	$(5.6 \pm 3.7) \times 10^{-19}$	1.0×10^{-6}	[23]

Table 2: Comparison of CSM physical parameters (using equipartition) with other supernovae where similar modelling methods have been used. The shock velocity has simply been calculated from $v = R/T$, where T is time since explosion and also some data was not constrained hence the blank entries. You can also see that some of the values were scaled to match our ϵ values and errors have been stated when published.

References

- [1] R. A. Chevalier, *ApJ*, **499**, 810 (1998).
- [2] L. DeMarchi *et al.*, *ApJ*, **938**, 84 (2022), arXiv:2203.07388.
- [3] M. Ugliano, H.-T. Janka, A. Marek, and A. Arcones, *The Astrophysical Journal* **757**, 69 (2012).
- [4] T. Shenar, Wolf-rayet stars, 2024, arXiv:2410.04436.
- [5] C. D. Kilpatrick *et al.*, *Monthly Notices of the Royal Astronomical Society* **480**, 2072 (2018), <https://academic.oup.com/mnras/article-pdf/480/2/2072/25440511/sty2022.pdf>.
- [6] D. Xiang *et al.*, *The Astrophysical Journal* **871**, 176 (2019).
- [7] P. Tadjimuratov and D. Mirzakulov, Sn2017ein: bolometric light curve and physical parameters, 2018, arXiv:1806.11288.
- [8] P. A. Crowther, *ARA&A*, **45**, 177 (2007), arXiv:astro-ph/0610356.
- [9] S. D. Van Dyk *et al.*, *ApJ*, **860**, 90 (2018), arXiv:1803.01050.
- [10] Y.-H. Zhao *et al.*, *The Astrophysical Journal Letters* **980**, L6 (2025).
- [11] J. A. Högbom, *A&AS*, **15**, 417 (1974).
- [12] I. Sushch, R. Brose, M. Pohl, P. Plotko, and S. Das, *The Astrophysical Journal* **926**, 140 (2022).
- [13] S. Xu and A. Lazarian, *The Astrophysical Journal* **850**, 126 (2017).
- [14] G. B. Rybicki and A. P. Lightman, *Radiative Processes in Astrophysics* (, 1986).
- [15] A. G. Pacholczyk, *Radio astrophysics. Non-thermal processes in galactic and extragalactic sources* (, 1970).
- [16] R. A. Chevalier and C. Fransson, *The Astrophysical Journal* **651**, 381 (2006).
- [17] B. W. Carroll and D. A. Ostlie, *An Introduction to Modern Astrophysics*, 2 ed. (Cambridge University Press, 2017).
- [18] K. D. Alexander, A. M. Soderberg, and L. B. Chomiuk, *The Astrophysical Journal* **806**, 106 (2015), arXiv:1405.0228.
- [19] S. Wellons, A. M. Soderberg, and R. A. Chevalier, *The Astrophysical Journal* **752**, 17 (2012), arXiv:1201.5120.
- [20] A. J. van der Horst *et al.*, *The Astrophysical Journal* **726**, 99 (2011), arXiv:1011.2521.

- [21] A. M. Soderberg, A. Brunthaler, E. Nakar, R. A. Chevalier, and M. F. Bietenholz, *The Astrophysical Journal* **725**, 922 (2010), arXiv:1005.1932.
- [22] A. J. Nayana, P. Chandra, A. Krishna, and G. C. Anupama, *The Astrophysical Journal* **934**, 186 (2022), arXiv:2206.12103.
- [23] A. J. Nayana, P. Chandra, and A. K. Ray, *The Astrophysical Journal* **863**, 163 (2018), arXiv:1808.00283.
- [24] K. Maeda *et al.*, *The Astrophysical Journal* **918**, 34 (2021), arXiv:2106.11618.
- [25] A. M. Soderberg *et al.*, *ApJ*, **621**, 908 (2005), arXiv:astro-ph/0410163.
- [26] R. A. Chevalier, *ApJ*, **258**, 790 (1982).
- [27] C. M. Stauffer *et al.*, *Monthly Notices of the Royal Astronomical Society* **505**, 1153–1161 (2021).
- [28] K. Maeda and T. J. Moriya, *The Astrophysical Journal* **927**, 25 (2022), arXiv:2201.00955.